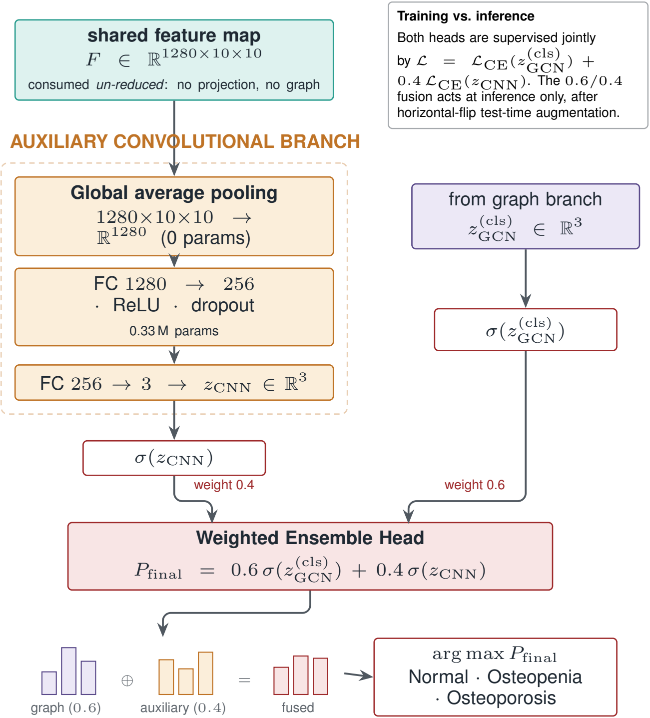

shared feature map
$F \in \mathbb{R}^{1280\times 10\times 10}$
consumed *un-reduced*: no projection, no graph
Training vs. inference
Both heads are supervised jointly by $\mathcal{L} = \mathcal{L}_{\mathrm{CE}}(z^{(\mathrm{cls})}_{\mathrm{GCN}}) + 0.4\,\mathcal{L}_{\mathrm{CE}}(z_{\mathrm{CNN}})$. The $0.6/0.4$ fusion acts at inference only, after horizontal-flip test-time augmentation.
AUXILIARY CONVOLUTIONAL BRANCH
Global average pooling
$1280\times 10\times 10 \rightarrow \mathbb{R}^{1280}$ (0 params)
FC $1280 \rightarrow 256$ · ReLU · dropout
0.33 M params
FC $256 \rightarrow 3 \rightarrow z_{\mathrm{CNN}} \in \mathbb{R}^3$
from graph branch
$z^{(\mathrm{cls})}_{\mathrm{GCN}} \in \mathbb{R}^3$
$\sigma(z^{(\mathrm{cls})}_{\mathrm{GCN}})$
$\sigma(z_{\mathrm{CNN}})$
weight 0.4
weight 0.6
Weighted Ensemble Head
$P_{\mathrm{final}} = 0.6\,\sigma(z^{(\mathrm{cls})}_{\mathrm{GCN}}) + 0.4\,\sigma(z_{\mathrm{CNN}})$
graph (0.6)
$\oplus$
auxiliary (0.4)
$=$
fused
$\arg\max P_{\mathrm{final}}$
Normal · Osteopenia · Osteoporosis